

E-Commerce Business Intelligence Dashboard

Business Intelligence & Analytics Report

Executive Summary

This project analyzes the Brazilian Olist marketplace dataset using PostgreSQL, SQL, Power BI and DAX. The analysis covers 99,441 orders, 96,096 unique customers and approximately \$16.01M in aggregated payment value. The strongest opportunities are customer retention, logistics optimization and understanding geographic and category concentration.

KPI	Result
Total orders	99,441
Unique customers	96,096
Total payment value	\$16.01M
Average order value	\$160.99
Repeat customer rate	3.12%
On-time delivery rate	91.88%
Late delivery rate	8.11%

1. Business Problem

The objective is to turn fragmented e-commerce transactions into decision-ready information. Management needs visibility into sales performance, customer behavior, category demand, delivery reliability, payments, reviews, sellers and regional demand.

2. Data & Methodology

The solution starts with CSV datasets containing orders, order items, payments, reviews, customers, products, sellers, geolocation and category translations. PostgreSQL is used for relational storage, constraints, indexes and analytical views. Power BI provides the semantic model, DAX measures and interactive reporting.

3. Sales Performance

Aggregated payment value is \$16,008,872.12, across 99,441 orders, giving an average order value of \$160.99. Revenue activity is concentrated in the 2017–2018 period. The dashboard's time-series views make monthly movement and seasonality easier to investigate.

4. Customer Analysis

There are 96,096 unique customers. Only 3.12% placed more than one order. This is a material retention opportunity: the marketplace is successful at acquiring customers, but repeat purchasing is comparatively limited.

5. Delivery & Logistics

Among delivered orders with the required dates, 91.88% were delivered on or before the estimated date and 8.11% were late. Logistics performance is therefore broadly strong, while the late-order segment remains large enough to justify regional and seller-level investigation.

6. Geographic Analysis

The top five customer states account for approximately 73.2% of total payment value. São Paulo is the largest contributor. This concentration suggests that service levels, seller coverage and logistics capacity in major states have an outsized impact on overall performance.

7. Category & Product Analysis

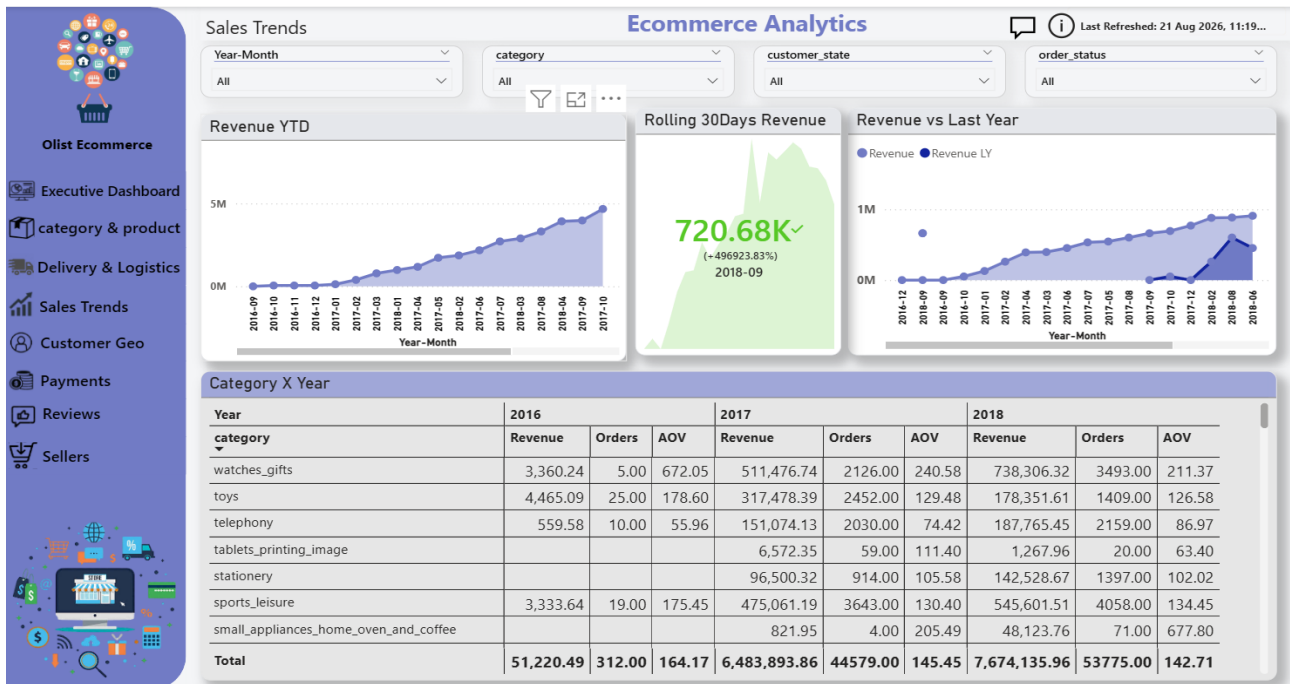
Using item-level product price as the category sales basis, the five largest categories contribute approximately 40.3% of product sales value. This indicates meaningful category concentration without implying that the entire business depends on only a handful of categories.

8. Dashboard & Analytical Views

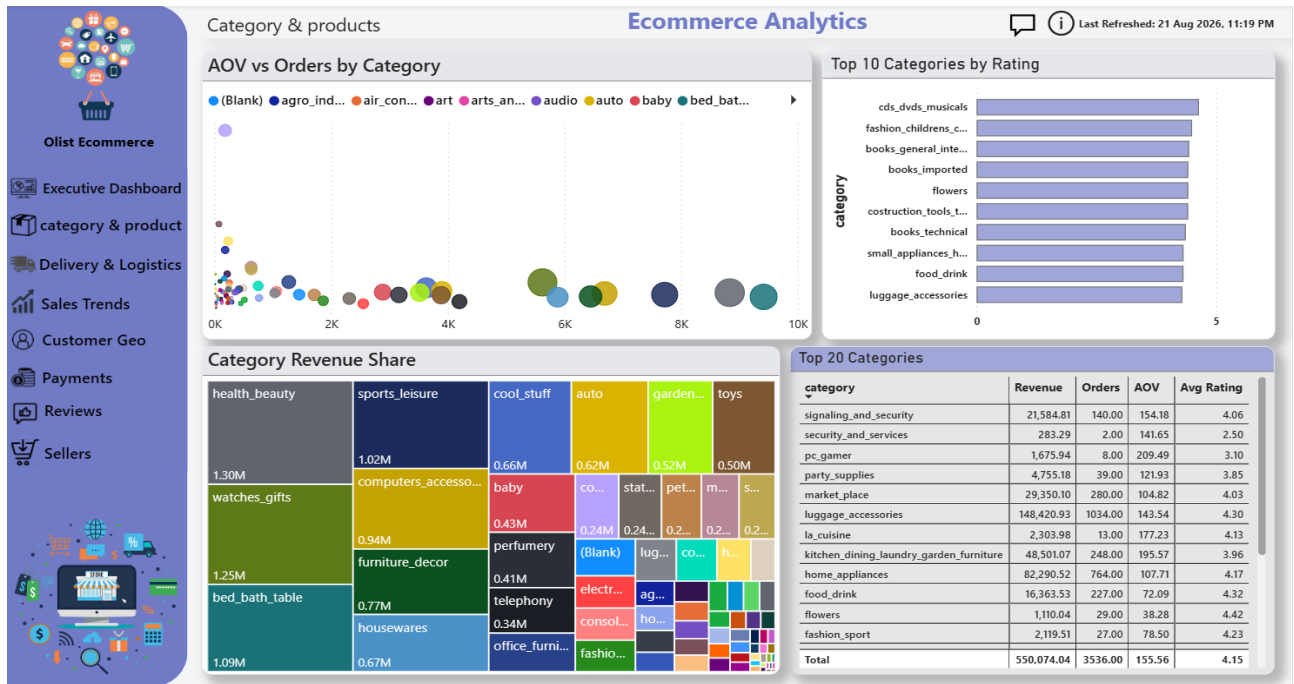
Executive overview with KPI monitoring and high-level trends.



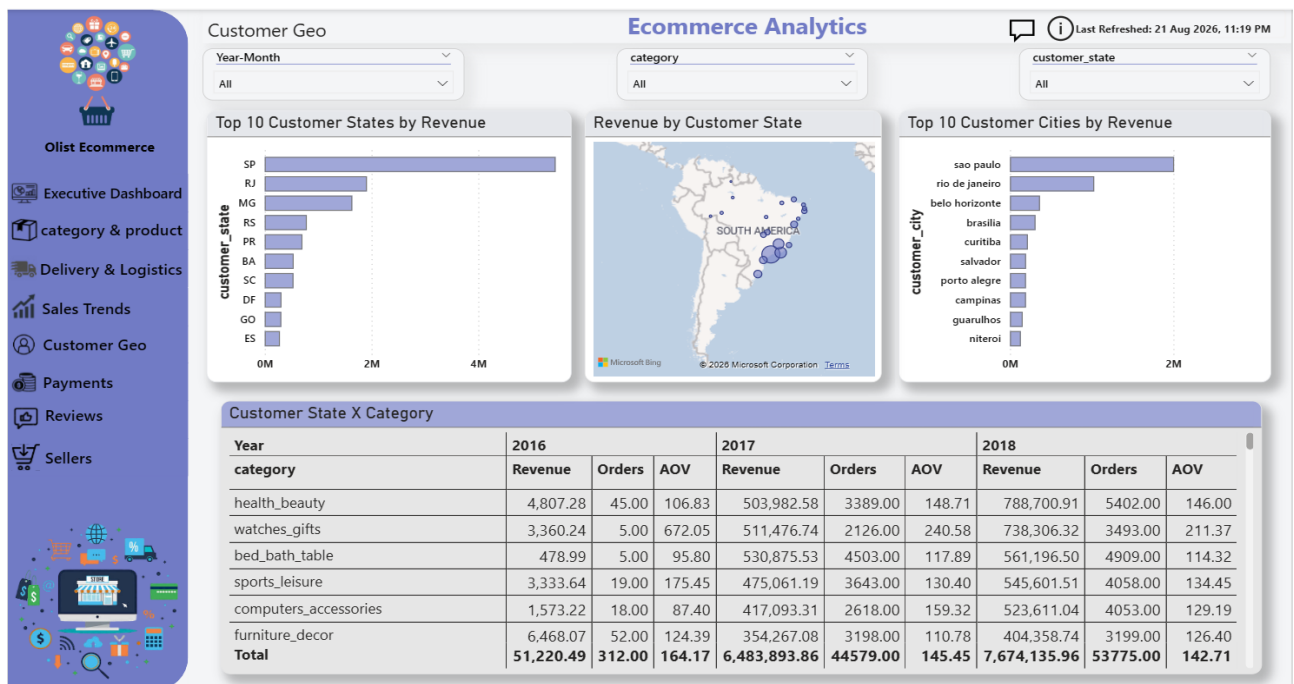
Sales trend analysis over time.



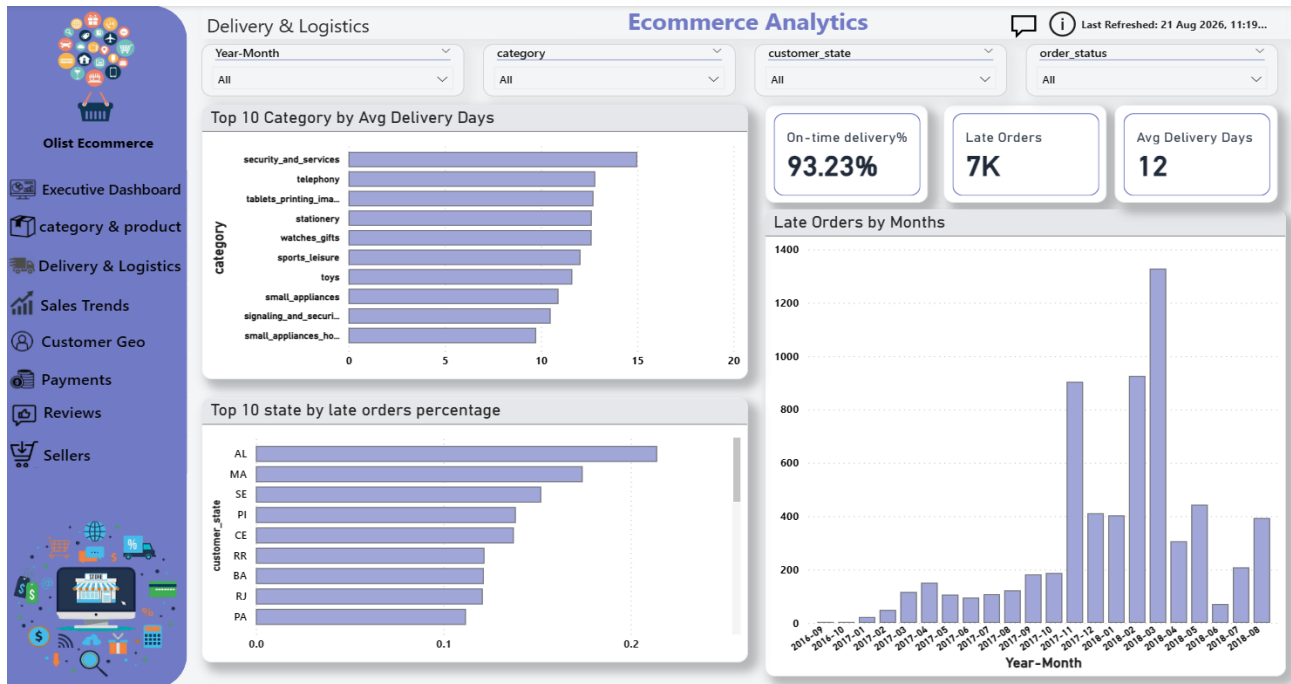
Category and product performance.



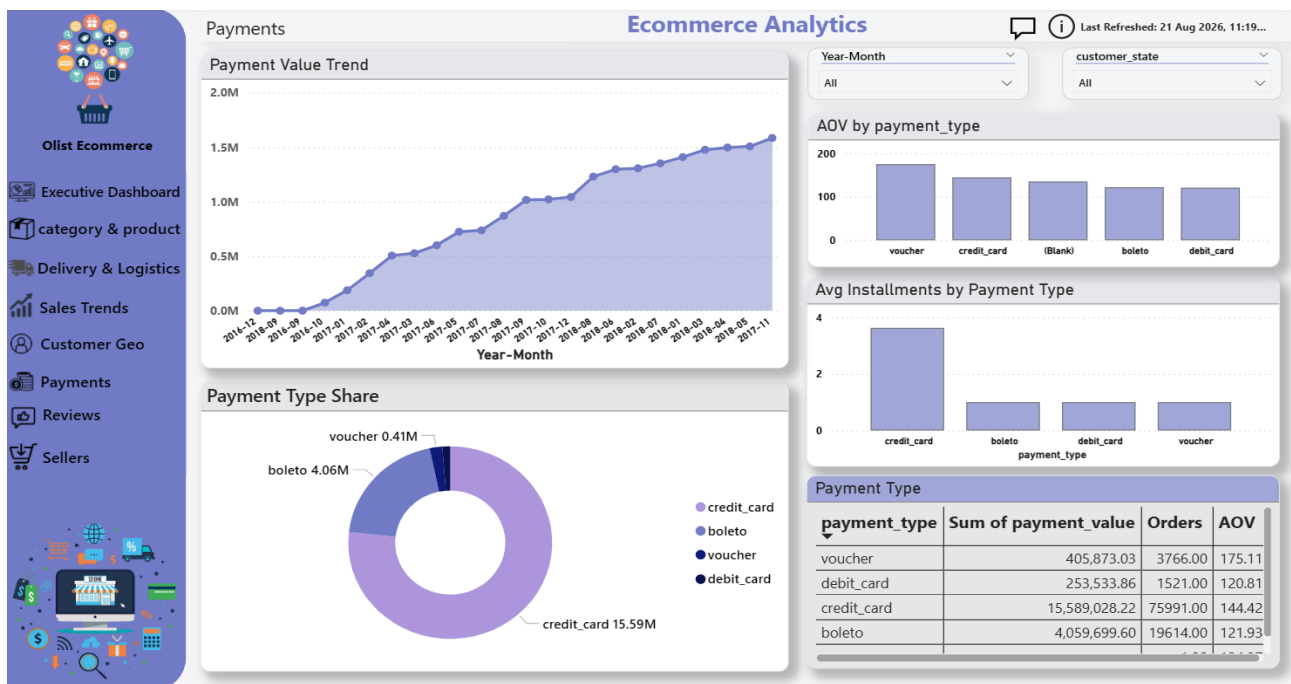
Customer and geographic distribution.



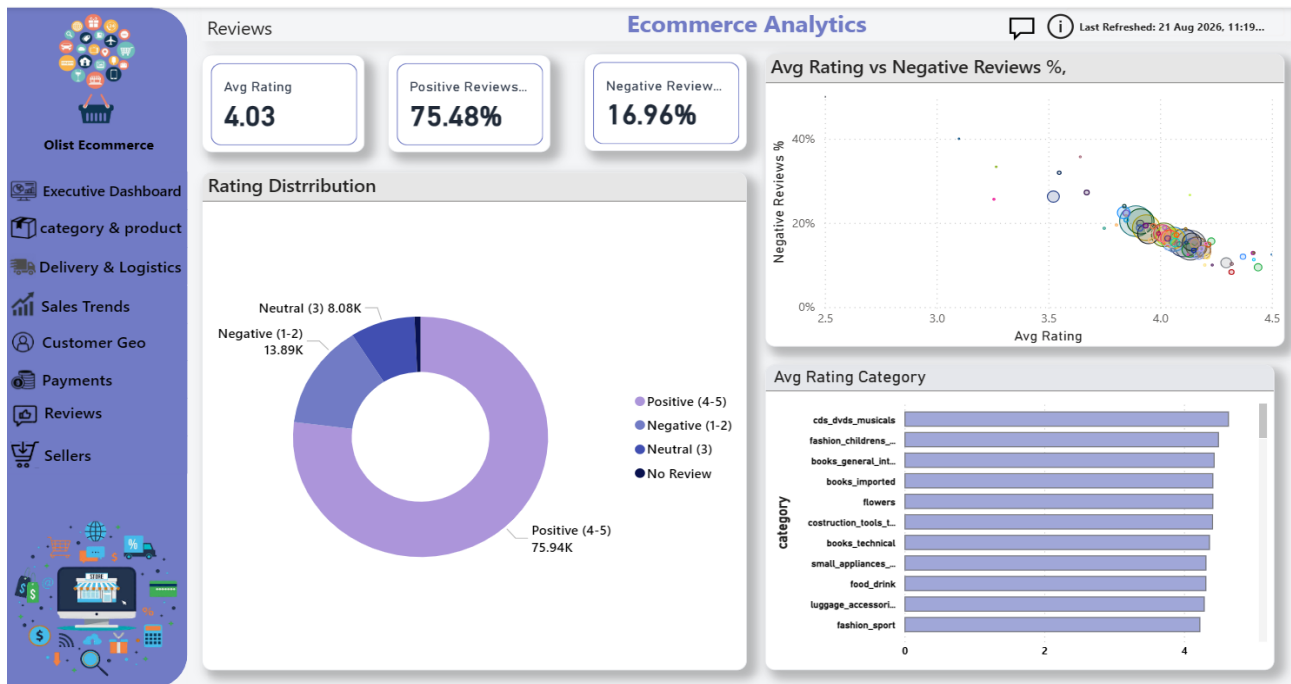
Delivery and logistics performance.



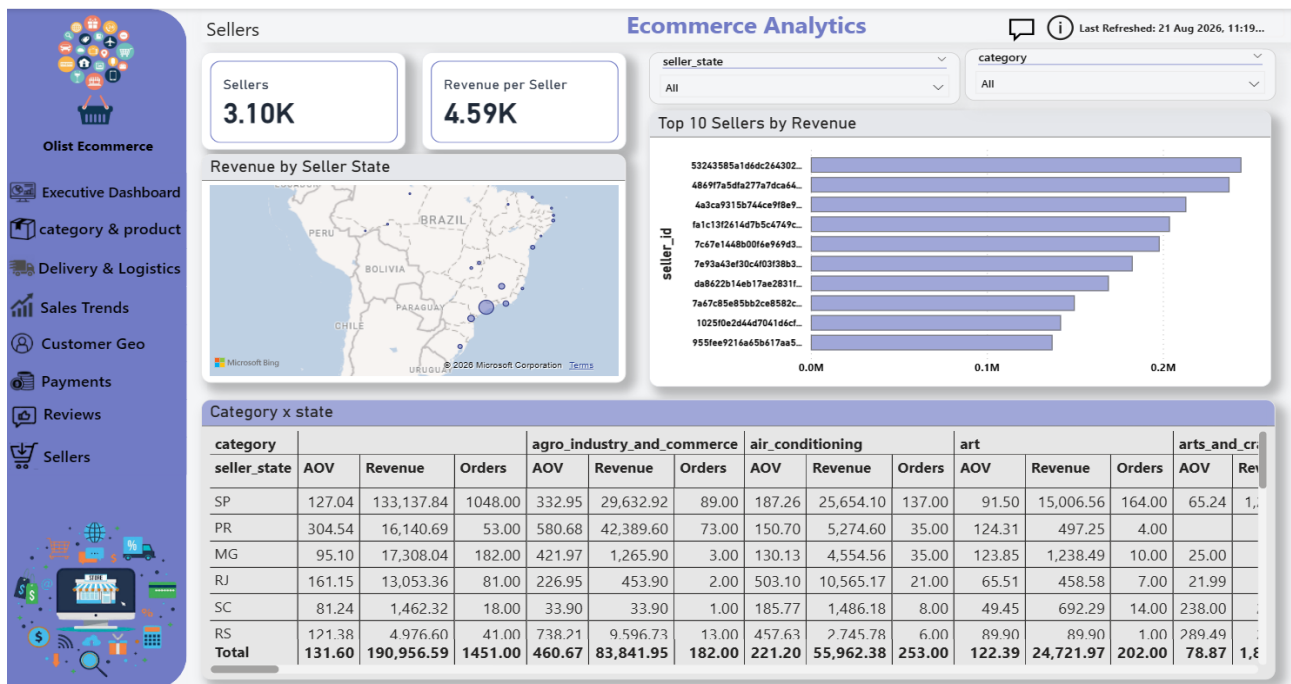
Payment behavior and transaction patterns.



Review and customer feedback analysis.



Seller performance analysis.



9. Strategic Recommendations

Retention

Create loyalty and repeat-purchase campaigns; segment customers by frequency and value; personalize offers.

Logistics

Investigate late delivery by state and seller; monitor delivery lead time and prioritize last-mile improvements.

Category Strategy

Protect availability of high-performing categories while testing pricing and promotion strategies for weaker categories.

Geographic Growth

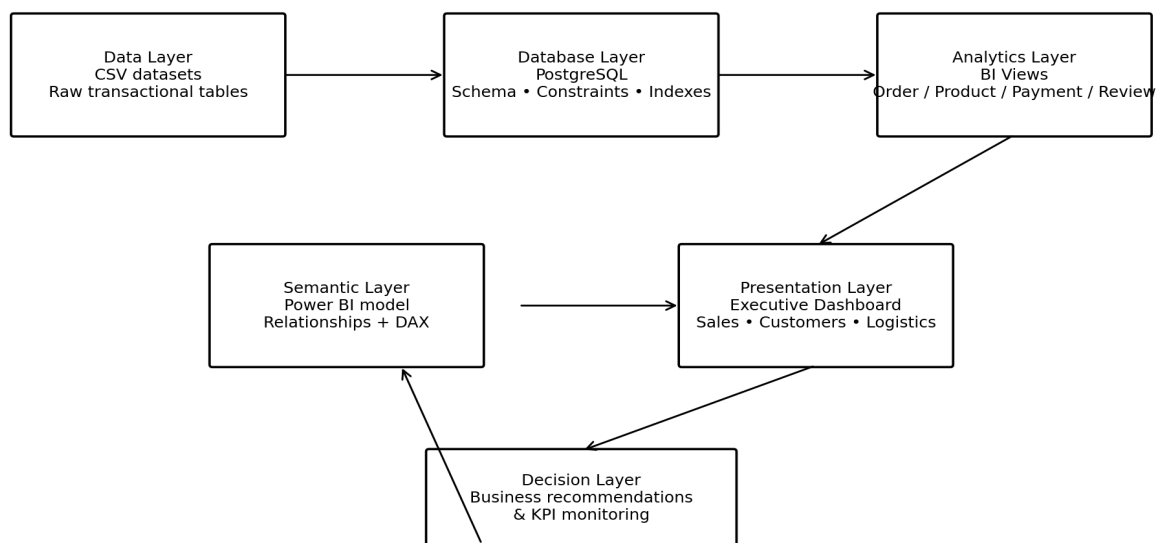
Maintain strong operations in high-revenue states and assess lower-revenue regions before increasing acquisition spend.

Measurement

Track revenue, AOV, repeat customer rate, on-time delivery, category mix and regional concentration as recurring management KPIs.

10. Technical Architecture

E-Commerce BI Solution Architecture



The solution separates data storage, SQL transformation, semantic modeling, visualization and decision support.

11. Limitations

The Olist dataset is historical and does not represent live marketplace conditions. Payment value, item sales value and revenue should be defined explicitly for each report. Delivery metrics require non-null delivery and estimated dates. The analysis is descriptive rather than predictive.

12. Conclusion

The dashboard demonstrates an end-to-end BI workflow from raw data to decision support. The most important business opportunities are improving repeat purchasing, reducing late deliveries and managing geographic/category

concentration through targeted operational and commercial actions.